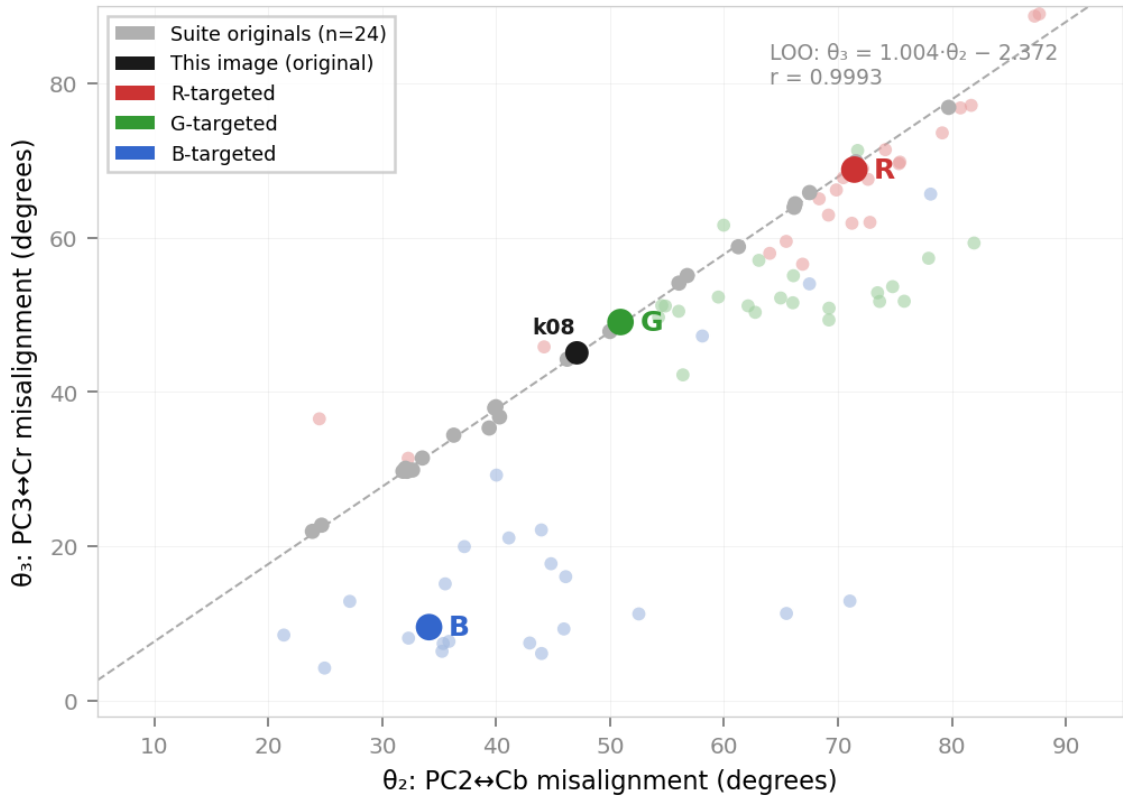


Compression Profile: KODIM08768x512 | 24-bit RGB | PNG (lossless)

market place | Strongly One-Dimensional (PC1 = 96.96%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9665	0.9162	0.9746	0.952	47.08°	45.09°	+0.19°
R-targeted	0.2472	0.2248	0.9191	0.464	71.44°	68.85°	-0.50°
G-targeted	0.1445	0.9078	0.1567	0.403	50.91°	49.08°	+0.34°
B-targeted	0.9803	0.0643	0.0231	0.356	34.1°	9.52°	-22.34°

R-targeted isolates red (R-G, R-B < 0.25) while preserving G-B coupling at 0.919.
G-targeted isolates green (R-G, G-B < 0.16) while preserving R-B at 0.908.
B-targeted preserves R-G at 0.980 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM08768x512 | 24-bit
RGB | PNG (lossless)

market place | Strongly One-Dimensional (PC1 = 96.96%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

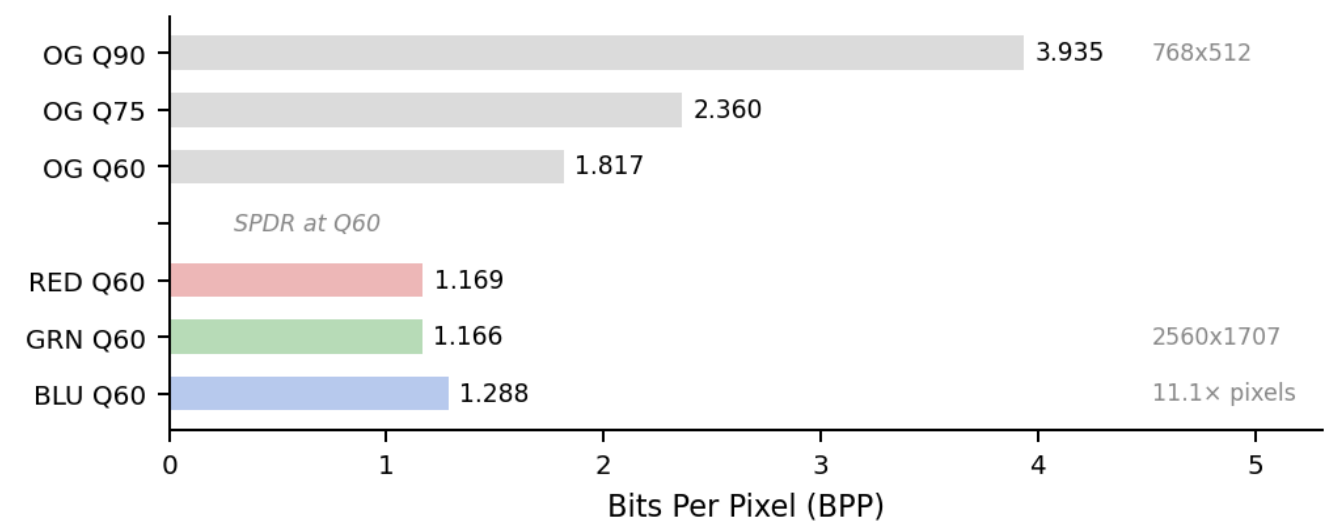
ORIGINAL (kodim08)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	770.0 KB	16.041	393,216
JPEG Q90	JPEG	768x512	188.9 KB	3.935	393,216
JPEG Q75	JPEG	768x512	113.3 KB	2.360	393,216
JPEG Q60	JPEG	768x512	87.2 KB	1.817	393,216

REDISTRIBUTED (kodim08 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	623.7 KB	1.169	4,369,920
GRN Q60	JPEG	2560x1707	621.8 KB	1.166	4,369,920
BLU Q60	JPEG	2560x1707	687.1 KB	1.288	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM08768x512 | 24-bit
RGB | PNG (lossless)

market place | Strongly One-Dimensional (PC1 = 96.96%)

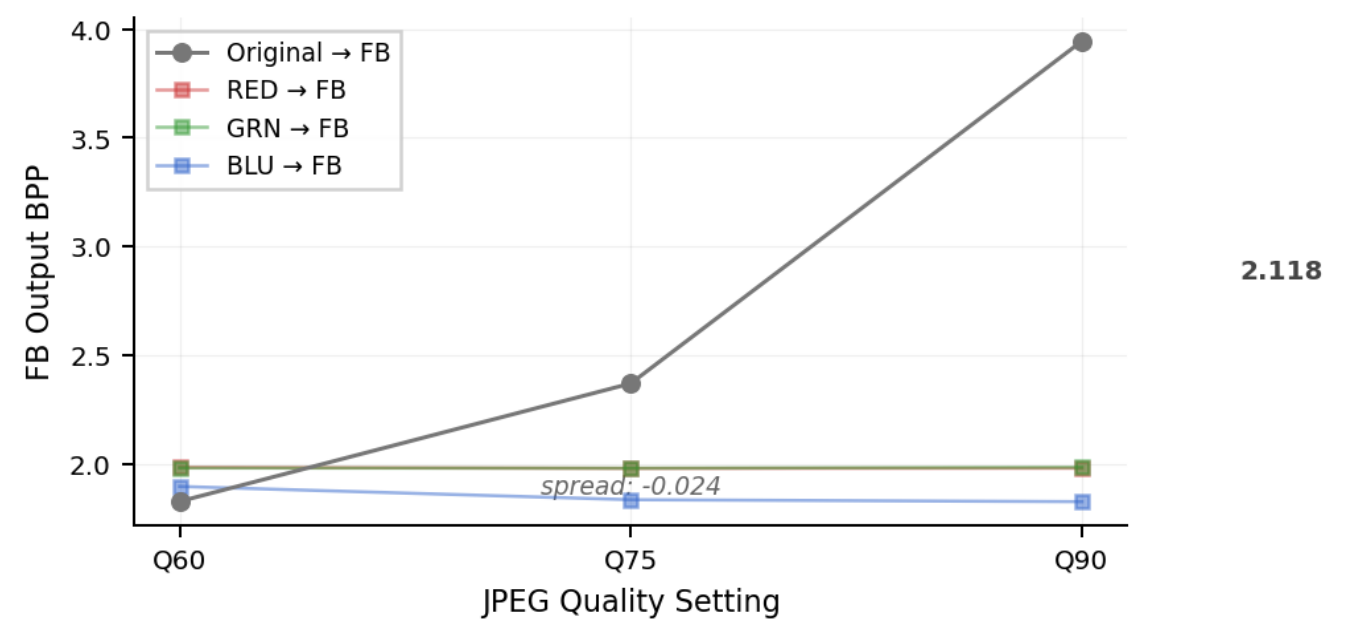
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	3.704	—
R-targeted → FB	2560x1707	2048x1366	1.980	46.5%
G-targeted → FB	2560x1707	2048x1366	1.984	46.4%
B-targeted → FB	2560x1707	2048x1366	1.826	50.7%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.417	61.7%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9517 vs 0.952 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 2.118 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.024 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim08

Best BPP (Q60, direct)	1.166	2560x1707, 4,369,920 px
Best BPP through FB	1.826	2048x1366 output
vs OG Q90 → FB	53.7%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216